

JUDE WATIMONGO

+233 548376114 / +256751332757 | watimongojude@gmail.com | watimongo.jude@ashesi.edu.gh
<https://www.linkedin.com/in/judewatimongo/>

PROFILE SUMMARY

Highly driven Master's student in Mechatronics Engineering with a strong foundation in hardware development and embedded systems, particularly in innovative prototype design. Passionate about Industrial Automation, energy management, and renewable energy, with hands-on experience in Industrial Automation and Energy Management. An active IEEE member with strong problem-solving skills, ready to apply technical knowledge and hands-on experience to impactful projects.

EDUCATION

Ashesi University

MPhil. Mechatronics Engineering

Cumulative GPA: 3.67/4.00

Relevant Courses: Control Systems, Robotics, Machine Learning, Energy Systems and Mobility, Industrial Automation

Berekuso, Eastern Region

Expected Date of Graduation: June 2027

Ashesi University

BSc. Electrical and Electronics

Cumulative GPA: 3.34/4.00

Relevant Courses: System Dynamics, Leadership, Programming, Circuit Design, Fundamentals of Design and Entrepreneurship

Berekuso, Eastern Region
Date of Graduation: June 2023

WORK EXPERIENCE

Dojo Hub, Uganda

Aug 2026 – Present

- Designed and implemented the electrical wiring and operational functions of an automated manufacturing line for bottle filling, capping, and tightening.
- Designed and programmed the Siemens S7 1214 PLC program to seamlessly handle the operations of the manufacturing line and also visualise the process on the Siemens KTP 400 HMI
- Diagnosed and resolved technical issues encountered during the installation, integration, and commissioning of the machines.

Sunbird AI, Uganda

June 2022 – Aug 2022

Hardware intern

- Developed prototype noise monitoring sensors utilising microcontrollers for effective environmental assessment.
- Implemented a voice feedback recording device prototype for an NGO called SEMA, enhancing public service feedback.
- Conducted rigorous testing of embedded systems to validate performance against established metrics in measuring sound pollution and calibrating the sensors.
- Presented a comprehensive account of the final solution to the board, resulting in approval and further project development.

PROJECT AND RESEARCH

Hybrid Electric Vehicle Energy Management Project

June - July 2025

Energy Optimisation project

- Designed and implemented an optimal EMS for a P2 hybrid vehicle using Pontryagin's Minimum Principle and ECMS, enabling real-time optimisation of engine-motor power split to reduce fuel consumption and maintain battery SOC.
- Built a MATLAB/Simulink controller that improved SOC tracking through a PI-based adaptive equivalence factor, achieving accurate SOC control on WLTC and SORT cycles and demonstrating strong controller robustness.
- Delivered measurable performance outcomes, including fuel-efficiency improvements across driving cycles and validated EMS behaviour through multi-cycle simulation analysis.

CO-CURRICULAR ACTIVITIES

IEEE Professional Organisation, Ghana Section

Feb 2025 – present

Graduate Student Member

- Competed in the IEEE Code Xtreme competition, achieving a top 8 ranking among teams in Ghana, demonstrating strong coding and problem-solving skills.
- Engaged with the IEEE Robotics and Automation Society, enhancing my expertise and contributing to knowledge-sharing initiatives within the field.

Resident Coordinator, Ashesi University

Dec 2025 – Jan 2026

- Served as the first point of contact for 150 students and the hostel's management to meet all logistical needs on campus, gaining a rating of 3.82/5.
- Liaised with service providers to resolve conflicts of interest with student interests within 2 days, improving living conditions on campus.

TECHNICAL SKILLS

- Programming Languages: Python, C/C++, MATLAB, Java
- Software/Tools: SolidWorks, Simulink, Arduino, PLC Programming, Proteus circuit design

REFERENCES

Available upon request